

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	N/A
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to develop an **AI & Machine Learning-based Rising Waters Flood Prediction System** that predicts flood risks using rainfall, river water levels, weather conditions, and historical flood data. The system provides early warnings to residents and disaster management authorities, helping reduce loss of life and property through timely alerts and informed decision-making.

Project Overview	
Objective	.The primary objective is to develop an intelligent flood prediction system using Artificial Intelligence and Machine Learning to accurately predict flood risks and provide real-time alerts to users and disaster management authorities.
Scope	The project focuses on collecting weather and river water-level data, preprocessing the data, applying machine learning algorithms for flood prediction, and displaying results through a web application with instant alert notifications.
Problem Statement	
Description	Floods caused by heavy rainfall and rising river water levels often result in significant damage to human life, agriculture, infrastructure, and property. Existing warning systems are often delayed or lack location-specific predictions, making timely evacuation difficult.
Impact	An AI-based flood prediction system will improve early warning capabilities, reduce disaster-related losses, support emergency planning, and help authorities take preventive measures before flooding occurs.
Proposed Solution	
Approach	Develop an AI & Machine Learning model that analyzes rainfall, river water levels, humidity, temperature, and historical flood records to predict flood risk levels. The system will continuously monitor incoming data and provide real-time flood predictions and alerts.

Key Features	Develop an AI & Machine Learning model that analyzes rainfall, river water levels, humidity, temperature, and historical flood records to predict flood risk levels. The system will continuously monitor incoming data and provide real-time flood predictions and alerts.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computing Resources	Intel Core i5/i7 or equivalent
Memory	RAM specifications	8 GB or above
Storage	Disk space for data, models, and logs	256 GB SSD or above
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv